

Probabilistic Battery Cycle-Life Prediction & Degradation-Aware Fast-Charging Optimisation

Complete Project Guide - what it is, how to use it, how it was trained, how reliable it is, and what it offers a power-electronics research group

Live dashboard	https://rajtheman-battery-fastcharging-dashboard.hf.space
Source repository	https://github.com/TheDeveloperAAA/battery-fastcharging-ml
Primary dataset	Toyota Research Institute / Severson et al. (Nature Energy 2019) + Attia et al. (Nature 2020) - public
Validation datasets	CALCE (Univ. of Maryland), NASA PCoE
Document generated	from results/ artifacts - every number in this PDF is read from the saved experiment outputs at build time

1. Executive summary

This project is a complete, deployed machine-learning system that answers two questions every fast-charging system must face:

(1) How long will this battery last? - predicted from only the first 100 charge-discharge cycles (~13% of the median cell's 782-cycle life; cohort lives span 148-2237 cycles), with a **calibrated 90% confidence interval** around every prediction, not just a point estimate.

(2) How fast can we charge without breaking a lifetime promise? - a risk-constrained optimiser that finds the fastest two-step constant-current charging policy whose 90%-confidence *lower* life bound still clears a required cycle-life target, plus the full speed-versus-life Pareto frontier.

Headline outcomes (all from `results/metrics.json` and `calibration_report.json`):

Quantity	Value	Reference point
Single-feature baseline accuracy (RMSE, primary / secondary test)	136 / 211 cycles	138 / 196 cycles published (Severson 2019, Table 1)
Best point predictor (5-fold conformal ensemble) RMSE	128 / 223 cycles	full-model benchmark: 100 / 214 cycles
90% interval coverage (PICP)	0.93 / 0.95	nominal 0.90 - intervals mean what they say
Mean 90% interval width	339 / 585 cycles	-
Recommended protocol @ L_min=800 cycles, 90% conf.	4.53C(0-67% SOC) then 3.29C to 80%	11.2 min to 80% SOC; guaranteed lower bound 898 cycles
Deployable estimator	120-tree LightGBM, 68 kB	primary-test MAPE 10.5% - BMS/DSP class

Everything is reproducible from a fresh clone with public data, is covered by 23 automated tests, and is live as an interactive dashboard (link on the cover).

2. What the system covers

A single predict-then-decide pipeline in four stages:

Stage	What it does	Key artifacts
0 - Data foundation	Downloads (resumable, size-verified, mirror fallback) and preprocesses three public datasets via Microsoft BatteryML's format; engineers the Severson early-cycle features; reproduces the canonical train / primary-test / secondary-test splits (41 / 42 / 40 cells)	<code>src/data/</code> , data summaries
1 - Probabilistic predictor	Elastic-net baselines, Optuna-tuned LightGBM, a two-branch 1D-CNN on the discharge-curve data, a stacked ensemble; conformal prediction (MAPIE) for calibrated intervals; compact estimator export; early-prediction and cross-dataset studies	<code>results/metrics.json</code> , <code>calibration_report.json</code> , <code>models/</code>
2 - Charging decision	Protocol-to-life surrogate + split conformal; minimise analytic 0-to-80%-SOC charge time subject to the 90%-confidence lower life bound clearing L_min; Optuna TPE search certified by dense grid; Pareto frontier	<code>results/protocol_recommendation.json</code> , <code>pareto_frontier.csv</code>
3 - Communication	Streamlit dashboard (5 views) on Hugging Face Spaces; SHAP global + per-cell explanations; publication figures; research note for a power-electronics audience	<code>app/</code> , <code>results/figures/</code> , <code>RESEARCH_NOTE.md</code>

Datasets actually processed in this build (`results/data_summary_*.json`):

Dataset	Cells processed	Cycle-life range	Role
TRI / MATR (Severson + Attia)	180 = 124-cell canonical cohort + 45 closed-loop cells + 11 outlier/noisy cells excluded from the benchmark splits (~10.2 GB raw)	148-2237 cycles (median 782)	training + both held-out test sets
CALCE CS2/CX2 (LCO prismatic)	13	448-828 cycles	cross-chemistry transfer test
NASA PCoE (B0005/06/07/18)	4	100-168 cycles (70%-of-rated EOL convention)	cross-dataset transfer test

Scope boundary, stated plainly: this is **battery-side prediction and charging-reference logic**. It does not design converters, switch semiconductors, or close current loops - it computes what the converter's outer loop should aim for.

3. How to use it (three user paths)

3.1 Stakeholder - open the dashboard

Open <https://rajtheman-battery-fastcharging-dashboard.hf.space> (free-tier Space; first load after long idle takes about a minute). Five views:

View	What you see / do
Cycle-life predictions	Predicted vs true life for every held-out cell with its 90% band; RMSE / MAPE / coverage metric cards; a reliability expander showing empirical vs nominal coverage
Charging-protocol advisor	Three sliders (C1, switch-point Q1, C2). Instant charge time, predicted life, 90% lower bound, and a pass/fail badge against the 800-cycle lifetime guarantee
Speed-vs-life Pareto frontier	The optimised frontier over the 72 lab-tested protocols (grey dots = observed protocols); table of recommended setpoints per life requirement
Explainability (SHAP)	Global feature importance, beeswarm, and a per-cell waterfall explaining any individual prediction
Live compact model	The actual 68 kB on-controller estimator running live: pick a real cell, perturb its degradation signals, watch the prediction respond

3.2 Researcher - reproduce from scratch

Requires Python 3.11, ~15 GB disk, internet; on macOS brew install libomp first. The README carries the exact copy-paste sequence; in outline: clone, create a venv, pip install -r requirements.txt, install BatteryML from source (it is not on PyPI), then run the numbered pipeline modules:

```
src.data.download → src.data.build_dataset → src.models.train →  
src.optim.protocol_opt → src.eval.explain → src.eval.figures. Every stage is idempotent  
(cached downloads and preprocessing are skipped) and configured from config.yaml (seed, alpha, L_min,  
protocol bounds). TRAINING_MODE=fast skips the 120-trial Optuna search for a quick run.
```

3.3 Controller engineer - consume the artifacts

Two files are designed to leave the repository: **models/compact_soh_estimator.txt** - a 120-tree LightGBM text dump (68 kB, max depth 3, <= 7 leaves per tree) that any C tree-walk or LightGBM's C API can execute in microseconds on a BMS/DSP-class processor; and **app/artifacts/protocol_grid.npz** - a precomputed 36 x 24 x 36 grid over (C1, Q1, C2) holding the surrogate's median life prediction, 90% lower/upper bounds, and charge time, so a supervisory controller can look up guaranteed-life charging setpoints with zero model inference.

4. Important features

Feature	Detail
Calibrated uncertainty, not vibes	Split-conformal AND 5-fold cross-conformal (MAPIE 1.4.1) intervals; empirical coverage is verified on two held-out test sets and reported next to nominal
Early prediction	Useful accuracy from the first 100 cycles; the error-vs-horizon study quantifies exactly what each extra observed cycle buys (Section 6, Figure 3)
Risk-constrained decision layer	The optimiser treats battery life as a chance constraint (CVaR-style): charge time is minimised only over protocols whose 90% lower life bound clears the target
Certified optimisation	Bayesian (Optuna TPE, seeded) search is cross-checked by a dense grid over the whole 3-D protocol space - the recommendation is the certified global optimum within grid resolution
Deployability	68 kB tree-ensemble estimator + a zero-inference protocol lookup grid; the dashboard itself runs on a free 2-vCPU Space
Explainability	Exact TreeSHAP attributions, global and per-cell; the top features are the physically expected degradation signals

Feature	Detail
Reproducibility	Pinned environment, global seeds, config-driven, resumable size-verified downloads with a Hugging Face mirror fallback, 23 automated tests, full decision log (BUILD_LOG.md)

5. How it is trained (precise)

5.1 Data and labels

Raw data: 8 channels per cell per cycle (voltage, current, temperature, charge/discharge capacity, time, internal resistance, and the dataset's precomputed 1000-point discharge-capacity-vs-voltage curve $Q_d(V)$ on 2.0-3.5 V). Label: $\log_{10}(\text{cycle life})$, cycle life = cycles until capacity falls to 80% of the 1.1 Ah rating, taken from the dataset's official per-cell field; our independent threshold-crossing labeler agrees with it to within 1 cycle on all 88 comparable cells (cross-check stored in `data_summary_matr.json`).

5.2 Features (Severson et al. 2019, verified against both the paper and BatteryML's implementation)

The core signal is $dQ(V) = Q_{d,\text{cycle-100}}(V)$ minus $Q_{d,\text{cycle-10}}(V)$: how the discharge curve has deformed over the first 100 cycles. From it: $\log_{10}|\text{variance}|$, $\log_{10}|\text{minimum}|$, skewness, kurtosis. Correlation of the variance feature with log cycle life on the 124-cell cohort: **-0.91** (paper reports -0.93). Plus: capacity-fade slope/intercept (cycles 2-100 and 91-100), early discharge capacity, average charge time (cycles 2-6), temperature integral, internal-resistance minimum and trend, and the charging-protocol parameters (C1, Q1, C2, per-SOC-window rates, analytic charge time).

5.3 Splits - leakage discipline

Canonical Severson splits, byte-identical to BatteryML's lists (unit-tested): 41 training cells, 42 primary-test cells (even/odd interleave of batches 1-2; outlier b2c1 excluded exactly as the paper's parenthetical metrics do), 40 secondary-test cells (the later batch 3 - a harder, genuinely out-of-batch test). The conformal calibration set (14 cells) is carved from the training cells only; feature scaling is fit on training data only; the test sets are touched once, at final evaluation.

5.4 Models

Component	Training detail
Elastic net (3 variants)	On the paper's variance / discharge / full feature subsets; l1-ratio and alpha by internal CV (Severson's own model family - the comparability baseline)
LightGBM	120-trial Optuna TPE (seeded) over a deliberately conservative space (≤ 15 leaves, depth ≤ 5 , strong L1/L2) - 41 training points overfit instantly otherwise; 5-fold CV objective; <code>n_estimators</code> = median early-stopped iteration across folds
Sequence CNN	Two-branch 1D-CNN: 3 x 1000 voltage-grid channels (Q_d at cycle 10, cycle 100, and their difference) and 3 x 100 per-cycle channels (fade, internal resistance, charge time); SmoothL1 loss, Adam, ReduceLROnPlateau, early stopping on an internal validation split; CPU, fully seeded
Stacked ensemble	Non-negative least squares on out-of-fold predictions, weights sum to 1; learned weights in this build: elastic-net-full 0.75, LightGBM 0.00, CNN 0.25 (NNLS zeroed the LightGBM - it duplicates the elastic net's signal with more variance; the data decided, not us)

5.5 Uncertainty calibration

MAPIE 1.4.1 (current v1 API). Two leakage-free variants: **split-conformal** (ensemble fit on 27 cells, absolute-residual quantile from the 14 held-out calibration cells) and **cross-conformal** ('CV-plus', 5 folds over all 41 training cells - the statistically efficient choice at this sample size, and the headline method). Intervals are formed on the log scale and exponentiated.

5.6 The decision layer's own model

A separate protocol-to-life surrogate (a hypothetical protocol has no early-cycle data, so the predictor above cannot be used directly). Inputs: deterministic transforms of (C1, Q1, C2). Family selected by 5-fold CV - quadratic ridge 289, GP-Matern 284, small LightGBM 254 cycles RMSE - LightGBM selected; fit on 74 cells, split-conformal on 49 held-out cells. The optimisation domain is clamped to the per-parameter ranges actually

spanned by the 72 tested protocols (C1 in [1, 8]C - the dataset includes 1C-first-step policies - Q1 in [5, 75]% SOC, C2 in [3, 6]C), so recommended setpoints never leave the tested range in any coordinate (box support; joint combinations between tested points rely on surrogate interpolation).

6. How reliable is it - verification, evidence, limits

6.1 Against the published benchmark (cited, not re-run)

Model	Ours: primary / secondary RMSE (cycles)	Severson 2019 Table 1 (excl. outlier)
Variance (1 feature)	136 / 211	138 / 196
Discharge (6 features)	314 / 141	86 / 173 (note: BatteryML's reproduction also degrades here, RMSE 329 - known skew/kurtosis instability)
Full (9 features)	138 / 232	100 / 214
Conformal ensemble (ours)	128 / 223	-

The single-feature baseline lands within 2 cycles of the published primary-test RMSE - strong evidence the data pipeline and feature engineering are faithful.

Early prediction from the first 100 cycles (ensemble + cross-conformal)

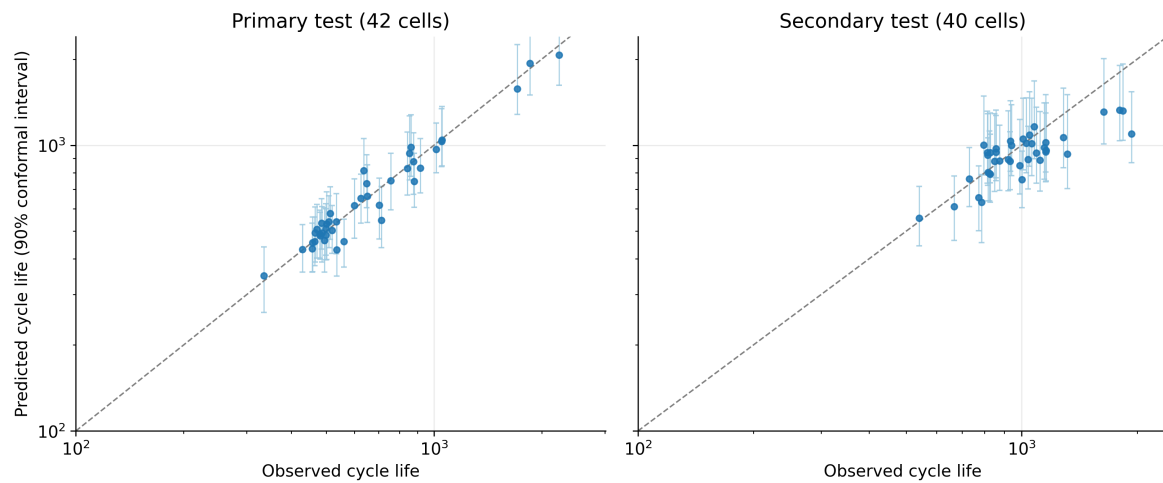


Figure 1 - predicted vs observed cycle life with 90% conformal intervals (left: primary test, right: secondary).

6.2 Calibration - the property the decision layer stands on

Method	Primary PICP	Secondary PICP	Primary MPIW	Secondary MPIW
cross-conformal (headline)	0.93	0.95	339 cyc	585 cyc
split-conformal	0.95	1.00	763 cyc	1,096 cyc

Nominal level is 0.90: both methods cover at or slightly above nominal (the conservative direction). The split variant over-covers because 14 calibration residuals make a coarse quantile - reported anyway, honestly.

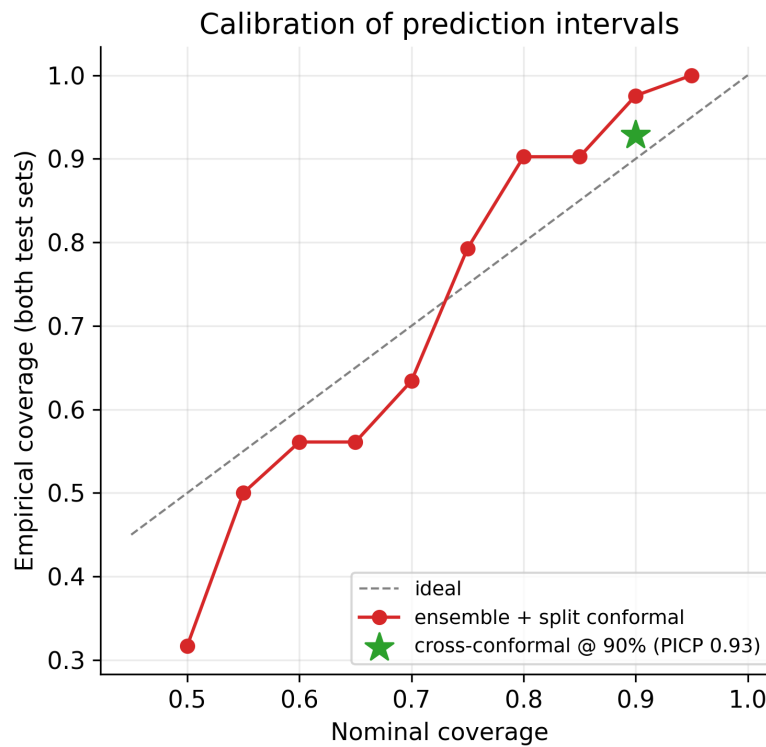


Figure 2 - empirical vs nominal coverage across confidence levels.

6.3 Early-prediction reliability

Error degrades gracefully as fewer cycles are observed (LightGBM, primary-test MAPE): 20 cyc: 15.1%, 30 cyc: 13.6%, 40 cyc: 12.9%, 50 cyc: 11.5%, 60 cyc: 12.8%, 70 cyc: 13.8%, 80 cyc: 12.7%, 90 cyc: 11.5%, 100 cyc: 10.8%.

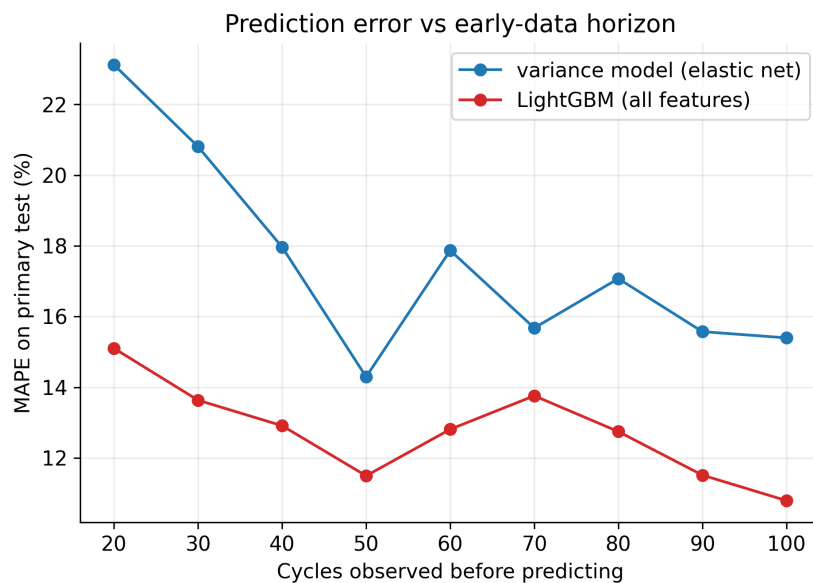


Figure 3 - prediction error vs number of observed cycles.

6.4 Software verification

23 automated tests: split lists are asserted byte-identical to BatteryML's source; feature definitions are validated on synthetic cells with known ground truth (including a no-future-leakage test that truncates the record and demands identical features); model determinism under fixed seeds; conformal coverage sanity on synthetic data; and all five dashboard views are executed end-to-end against the real artifacts via Streamlit's official AppTest harness. Downloads are size-verified to the byte against the upstream Content-Length values.

6.5 Known limits (read before trusting it blindly)

Chemistry transfer largely fails zero-shot, and we say so. NASA PCoE: rank correlation 1.00 (perfect ordering, $n=3$) but about 65% absolute scale error. CALCE (LCO prismatic, gentle cycling): Spearman -0.10 - the LFP fast-charge degradation signature does not rank LCO cells at all. Applying this model to a new chemistry requires recalibration data. **Conformal guarantees are marginal** (on average over cells like the training population), not per-cell. **The optimiser is only valid inside the observed protocol envelope** - it is clamped there by construction. All cells were cycled at 30 C chamber temperature; thermal extrapolation is untested.

7. Pros and cons

Pros	Cons
Uncertainty you can act on: coverage-verified 90% intervals feed a chance constraint, which is the technically correct way to make speed-vs-life decisions	Small training set (41 cells) - intervals are honest but not narrow; more cells would shrink them
Reproduces the published benchmark (within 2 cycles on the headline baseline) from a fresh clone of public data	Zero-shot chemistry transfer fails (CALCE) - LFP-specific; new chemistries need fine-tuning data
Decision output is directly actionable: two CC current setpoints + a switchover SOC, certified optimal within the tested protocol family	Protocol space is the TRI two-step family to 80% SOC at 30 C; pulse/AC/thermally-coupled charging is out of scope
Deployable: 68 kB estimator, zero-inference protocol lookup grid, free-tier dashboard	Compact model gives point estimates only; intervals come from the (heavier) conformal ensemble
Fully open: public data, pinned environment, decision log, 23 tests, live demo	Cycle-life labels need the cell to be cycled to end-of-life once per protocol family - the method predicts early, but training data is expensive to create

8. What it offers the research group, interest by interest

Mapped against the group's stated interests. Honesty labels: **[direct]** = usable as-is; **[adjacent]** = same artifacts, one integration step away; **[methodological]** = the technique transfers, the trained model does not.

Research interest	What this project contributes
EV charging stations [direct]	The core deliverable. Health-aware charging references: the recommended 4.53C/3.29C two-step policy and the whole Pareto table are CC-stage setpoint schedules with an explicit lifetime guarantee. Per-port life predictions with confidence bounds enable health-aware power allocation across a multiport station: ports holding cells near their life constraint get throttled first, quantified in cycles - not by heuristic
Modular multilevel / multilevel converters [adjacent]	For battery-integrated MMCs (split-battery submodules), the compact per-cell SoH/life estimator (68 kB; runs per submodule controller) provides the state input that submodule-level sorting / state-of-health balancing algorithms need; the conformal bound tells the balancer how much to trust it
Solid-state transformers [adjacent]	SST-based DC fast chargers need exactly the outer-loop current reference this system produces; the SoH-adaptive reference (the estimator re-evaluated as the cell ages) lets the same hardware hold a lifetime guarantee over the fleet's life
Grid integration of renewables [methodological + adjacent]	Battery energy storage scheduling faces the identical trade-off (cycle harder now vs preserve life). The predict-with-intervals → chance-constrained-optimisation pattern transfers one-to-one to degradation-aware BESS dispatch; the trained LFP model itself applies where LFP packs are used, with the cross-chemistry caveat of Section 6.5
HVDC / MVDC / FACTS [methodological]	No trained artifact applies directly - stated plainly. What transfers is the decision framework: risk-constrained operating-point optimisation against a learned degradation/reliability surrogate with calibrated uncertainty (e.g. loading vs insulation/semiconductor ageing), plus the compact-model-on-DSP deployment pattern proven here. Also relevant where BESS interfaces at MVDC nodes of a DC charging hub

Concrete collaboration hooks: (i) an SoH-adaptive CC-CV reference for the dual-active-bridge work - simulation-only integration is possible today by feeding the recommended current profile into a PLECS/Simulink DAB model; (ii) a SEFET/ECCE-style co-authored paper on predict-then-decide fast charging with calibrated uncertainty - the figures in results/figures/ are publication-format; (iii) a teaching example for digital-control courses: a real ML estimator small enough to step through on a DSP.

9. Why it should be used

Because a point estimate cannot make this decision. Charging faster always wins on time and always loses on life; the only defensible way to pick a protocol is to bound the life loss with stated confidence. This system is built around that bound: the interval is calibrated (Section 6.2), the constraint consumes the interval's lower edge, and the optimiser is certified against exhaustive search. The alternative - committing each candidate protocol to a year of cycling - is exactly what early prediction exists to avoid (predicting at cycle 100 instead of cycling to death saves ~87% of test time for the median 782-cycle cell, at 10.1% MAPE).

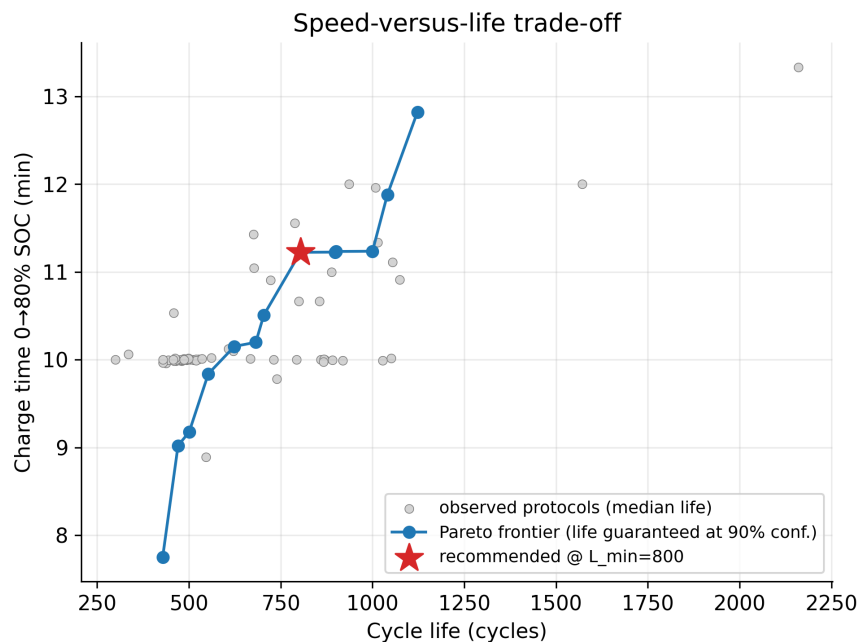


Figure 4 - the speed-versus-life frontier; the star is the certified recommendation at the 800-cycle / 90% guarantee.

The frontier spans 7.8-12.8 minutes (0-80% SOC) as the life requirement sweeps 400-1,100 cycles. The sweep was run to 1600 cycles; above 1,100 no protocol in the tested family satisfies the 90% bound - knowing where the guarantee becomes impossible is itself a result.

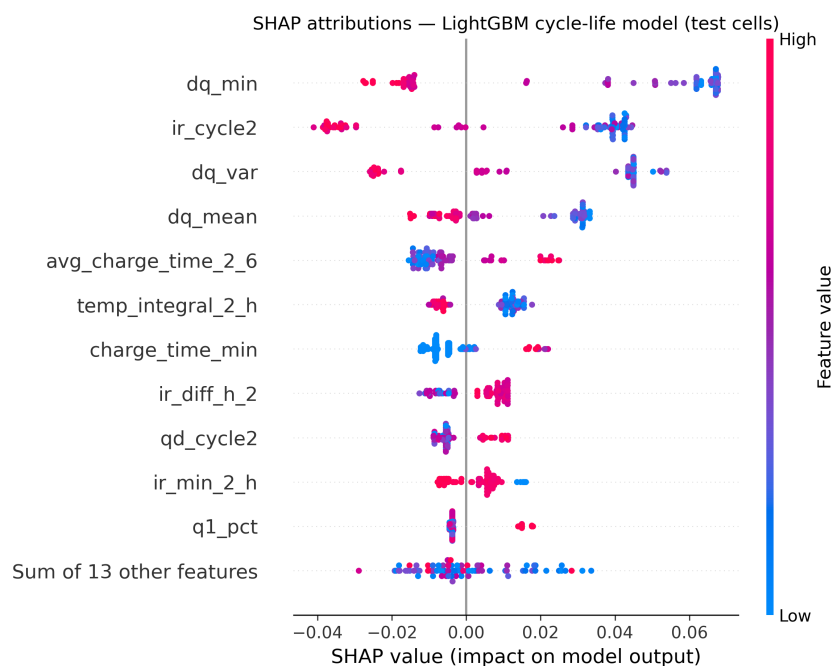


Figure 5 - SHAP attributions: the model's reasoning is dominated by the physically expected degradation signals (dQ(V) statistics, internal resistance).

10. Links, citation, licence

Live dashboard	https://rajtheman-battery-fastcharging-dashboard.hf.space
Repository (code, tests, BUILD_LOG)	https://github.com/TheDeveloperAAA/battery-fastcharging-ml
Walkthrough notebook	notebooks/walkthrough.ipynb (executed - outputs are real)
One-page note for the group	RESEARCH_NOTE.md
All metrics quoted here	results/metrics.json, results/calibration_report.json, results/protocol_recommendation.json, results/pareto_frontier.csv

Data and key references: K. A. Severson et al., Nature Energy 4, 383-391 (2019), doi:10.1038/s41560-019-0356-8. P. M. Attia et al., Nature 578, 397-402 (2020), doi:10.1038/s41586-020-1994-5. BatteryML (Microsoft, ICLR 2024). CALCE Battery Research Group, Univ. of Maryland. NASA Prognostics Center of Excellence Battery Data Set. All datasets are public; no laboratory hardware was used or required.

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